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(2)

Measures of Variations (Dispersions)

Mathematical Measures of Variations

The Standard Deviation and the Variance

The standard deviation is the most commonly used and the most important measure of variability. It uses the mean of the distribution as a reference point and measures variability by considering the *distance between each observation and the mean.*

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Standard deviation determines whether the scores are *near or far from the mean*. That is, are the scores *clustered or scattered?*

In simple terms, the standard deviation approximates *the average distance from the mean*.

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Standard deviation is defined for *population data* and *sample data*.

i) *Standard deviation and variance for population data*

Definition 1:- The population standard deviation, denoted by σ , is defined to be

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$$

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Definition 2:- The population variance , denoted by σ^2 is defined as the square of standard deviation.

$$\text{i.e } \sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$

Example 1:- Find the variance and standard deviation of the following population data.

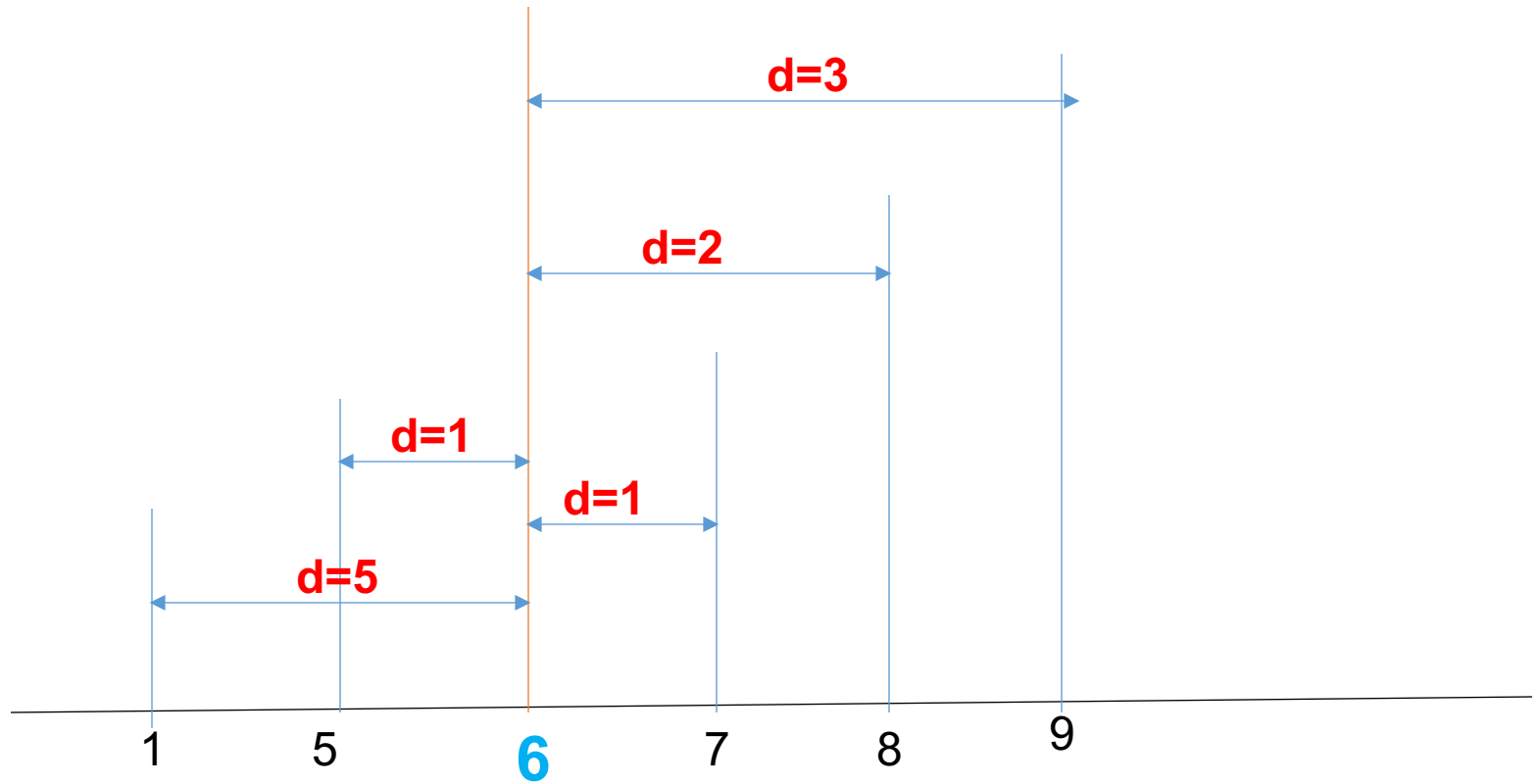
1, 9, 8, 7, 5

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Remember always that the purpose of variability is to determine how spread out the scores. Standard deviation accomplishes this by providing a measurement of standard distance from the mean.

Consider the population data given in example 1.

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ii) Standard deviation and variance for sample data

Definition 3:- The sample variance , denoted by s^2 is defined as the square of standard deviation.

$$\text{i.e. } s^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}$$

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Definition 4:- The sample standard deviation, denoted by s , is defined to be

$$s = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}}$$

The fact that a sample tends to be less variable than its population means that sample variability gives a biased estimation of population variability.

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To correct this bias, it is necessary to make an adjustment in the calculation. The effect of adjustment is to increase the value you obtain. Dividing by a smaller number (i.e. by $N - 1$ instead of N) produces a larger result and makes sample variance an accurate, or unbiased estimator of population variance.

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This is called Bessel's correction. Friedrich Wilhelm Bessel (German: 1784 – 1846) was a German astronomer, mathematician, physicist, and geodesist.

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By dividing by $n - 1$, we:

- ☐ Inflate the result slightly to **account for the fact that we're missing information** (the full population).
- ☐ Ensure that the sample standard deviation is an **unbiased estimator** of the population standard deviation.

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iii) Standard deviation and variance for GFD

In the case of GFD to find the sample standard deviation, we use the following formula:

$$s = \sqrt{\frac{\sum_{i=1}^k f_i (x_i - \bar{x})^2}{N - 1}},$$

where k – the number of classes of the GFD

f_i – the class frequency of the i^{th} class

x_i – the class mark of the i^{th} class

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Example 2:- Consider the GFD you have constructed in chapter 3 and find the standard deviation and the variance of the given data.

Weight (in Kg)	Number of students
35 - 43	8
44 - 52	11
53 - 61	16
62 - 70	19
71 - 79	14
80 - 88	9

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iv) *Properties of standard deviation*

Suppose the standard deviation of

$x_1, x_2, x_3, \dots, x_N$ is s . Then

✓ the standard deviation of

$x_1 + k, x_2 + k, x_3 + k, \dots, x_N + k$ is s (*unchanged*).

✓ the standard deviation of

$x_1 - k, x_2 - k, x_3 - k, \dots, x_N - k$ is s (*unchanged*).

✓ the standard deviation of

$x_1 \times k, x_2 \times k, x_3 \times k, \dots, x_N \times k$ is $s \times k$.

(*changed by k times the old standard deviation*)

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Example 3:- Consider the data 1, 9, 8, 5, 7

(i) Add 2 to each observation and find the standard deviation and compare this result with the original standard deviation.

(ii) Multiply each observation by 3 and find the standard deviation and compare this result with the original standard deviation.

Example 4:- The mean and standard deviation of the selling price in *Abadir Super Market* are 36 and 3.30 birr respectively. If the discount of 10% is made on each of the items, then find the mean and standard deviation after the discount.

Mathematical Measures of Variations

Standard deviation determines whether the scores are *near or far from the mean*. That is, are the scores clustered or scattered?

In simple terms, the standard deviation approximates *the average distance from the mean*.

Variance for inference and standard deviation for description

Standard deviation is one of the key methods that analysts, portfolio managers, and advisors use to determine risk. When the group of numbers is closer to the mean, the investment is less risky; when the group of numbers is further from the mean, the investment is of greater risk to a potential purchaser.

Variance for inference and standard deviation for description

Example 5:- We begin with a population that consists exactly $N = 6$ scores : 27, 27, 29, 29, 31, 31. With a few single calculations you should be able to verify that this population has a mean of

$$\mu = \frac{27(2) + 29(2) + 31(2)}{6} = 29,$$

*Variance for inference and standard deviation
for description*

and a variance of

$$\sigma^2 = \frac{2(27 - 29)^2 + 2(29 - 29)^2 + 2(31 - 29)^2}{6}$$

$$= \frac{16}{6} = 2.67 \text{ and a standard}$$

$$\text{deviation of } \sigma = \sqrt{2.67} = 1.63$$

Variance for inference and standard deviation for description

Next, we will select samples of $n = 2$ scores from this population. In fact we will obtain every simple possible sample with $n = 2$.

Variance for inference and standard deviation for description (Contd ...)

Sample	1 st score	2 nd score	SAMPLE STATISTIC		
			Mean ($\bar{x}$)	Variance (s^2)	Standard Deviation (s)
1	27	27			
2	27	29			
3	27	31			
4	29	27			
5	29	29			
6	29	31			
7	31	27			
8	31	29			
9	31	31			

Variance for inference and standard deviation for description (Contd ...)

Sample	1 st score	2 nd score	SAMPLE STATISTIC		
			Mean ($\bar{x}$)	Variance (s^2)	Standard Deviation (s)
1	27	27	27	0	0
2	27	29	28	2	1.41
3	27	31	29	8	2.83
4	29	27	28	2	1.41
5	29	29	29	0	0
6	29	31	30	2	1.41
7	31	27	29	8	2.83
8	31	29	30	2	1.41
9	31	31	31	0	0
			$\bar{x} = 29$	$\bar{s}^2 = 2.67$	$\bar{s} = 1.26$

Any Questions



Any Questions



Thank
you